Engineering Excellence in Motion
How IFCPL Became India's Valve Virtuoso

From a Chennai workshop to becoming the trusted guardian of India's thermal power infrastructure, Ideal Flow Control has mastered the art of keeping the nation's energy lifeline beating.



Dr Jegannathan Ganesan, Director of Ideal Flow Control Pvt Ltd, Chennai

In the unforgiving world of thermal power generation, where a single valve malfunction can trigger cascading shutdowns costing crores per day, Dr. G. Jegannathan emerged as India's most trusted problem-solver. As Managing Director of Ideal Flow Control Private Limited (IFCPL), he transformed a modest Chennai-based enterprise established in 2007 into an indispensable partner for the nation's energy sector—one critical valve at a time.

IFCPL's journey mirrors India's own industrial evolution: from dependency on foreign expertise to homegrown technical mastery. When Jegannathan, leveraging his extensive BHEL background, founded the company, he identified a critical gap—India's thermal power plants lacked specialized in-situ valve reconditioning services. Plants faced expensive shutdowns or relied on imported solutions for valve failures. IFCPL's answer was revolutionary: bring world-class valve surgery directly to the plant floor.



"Power plants cannot afford failures," Dr. Jegannathan explains. "Our in-situ valve reconditioning enables repairs without shutdowns, saving clients tens of crores while keeping electricity flowing to millions of homes and industries."

This philosophy drove IFCPL's technical evolution. The company developed capabilities for on-site machining of critical-access valve components—HP/LP Bypass valves, Main Steam Stop Valves, Turbine MAL Valves, Safety Relief Valves—without dismantling entire systems. Their pioneering Residual Life Assessment (RLA) studies enabled predictive maintenance, transforming valve management from reactive firefighting to strategic planning.

The numbers validate IFCPL's impact: over 10,000 valves refurbished across NTPC facilities, state electricity boards, and private power producers. Each refurbishment represents not just technical expertise but operational continuity for India's energy grid. During planned maintenance shutdowns and emergency breakdowns alike, IFCPL's specialized teams have become the first call for plant managers nationwide.

The COVID-19 pandemic proved IFCPL's resilience. While many industries faltered, IFCPL maintained essential services—because power generation couldn't stop. Adapting rigorous safety protocols, the company's engineers continued critical on-site work, ensuring India's lights stayed on during the nation's darkest hours.

Recognition followed achievement. In 2025 alone, Dr. Jegannathan received the Outstanding Achievement Award for Business Excellence from Minister Ramdas Athawale, the Nelson Mandela Leadership Award in Oxford, the ET Indo Global Leaders Award in Dubai, and most recently, the Global Innovation & Leadership Award for Outstanding & Most Innovative Engineering 2025. Fortune India and Business Today magazines honored him among their prestigious leadership lists.

"We don't manufacture valves—we extend their life, making existing infrastructure more efficient," Dr. Jegannathan notes. "This proves India's strength lies not just in building new but maintaining existing infrastructure brilliantly."

Under his stewardship, supported by directors Mythili Jegannathan, Dr. Jegannathan Srimathidevi, and Jegannathan Samyukthai, IFCPL exemplifies specialized technical excellence combined with innovative service delivery. Their 24/7 operational capability has become crucial for India's continuous industrial processes.

As India Today celebrates five decades of chronicling national progress, IFCPL's story represents an essential truth: India's industrial maturity requires not just large conglomerates but specialized service providers mastering complex niches—the unsung heroes keeping infrastructure running, one precision-engineered solution at a time.